

PhytoGraph — cycles 44-46

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PhytoGraph — cycles 44-46

Abstract

Cycles 44-46 preserved the terminal PhytoGraph closure state. The supplied audit report records no critical findings, no moderate findings, and no new minor findings. It validates the worker output as a terminal no-op: no defect, reproducibility failure, public-facing issue, or qualifying scientific evidence was supplied, so no build action, artifact rebuild, file edit, or ledger promotion was warranted during the worker/auditor cycle work.

The audit decision was `VALIDATED`. The master `prediction_ledger.tsv` and `speculation_ledger.tsv` each remained header-only with 1 line. `promise_ledger.jsonl` contained 237 events. `promise_check` exited 0 with 237 events and 46 plan milestones, and `org_check` exited 0 with inherited root-layout warnings only.

No track reopening predicate was met. All six tracks remain conservatively closed under the previously established limitation-class outcomes. The original research success criterion of at least one validated prediction per track remains unmet, and the campaign remains closed as a conservative infrastructure, validation, falsification, public communication, and handoff package.

Introduction

PhytoGraph’s controlling boundary is the master-ledger promotion rule. A track-local result, Atlas display, closure note, or maintenance artifact does not become a campaign-level prediction unless it is promoted into the master `prediction_ledger.tsv` under the campaign validation protocol. A claim that cannot name a validation source belongs in `speculation_ledger.tsv`; neither ledger received data rows during this cycle range.

Earlier substantive work established the terminal state. The Live Recovery Addendum directed continuation from the last substantive state represented by `reports/cycles/report_cycles_17-19.md` and post-merge integration artifacts for fork `cc044bf40be3`, while treating later no-work continuity reports as non-scientific. Subsequent closure reporting preserved the

six-track limitation statuses and the non-promotion boundary. Cycles 41-43 already recorded a validated no-op terminal maintenance window. Cycles 44-46 continue the same posture.

This report consolidates cycles 44-46. It does not re-audit the scientific results or reopen the tracks. Its role is to explain what happened in this reporting window, what was preserved, and what remains open.

Approach

The primary source for this report is the supplied cycles 44-46 audit report. Local workspace artifacts were used for continuity and simple sanity checks: MANIFEST.md, prediction_ledger.tsv, speculation_ledger.tsv, promise_ledger.jsonl, REFERENCES.md, reports/cycles/report_cycles_41-43.md, reports/final_campaign_handoff_manifest.md, reports/taxonomy_results_site_closure_note.md, and reports/taxonomy_results_site_qa.md.

The supplied cycle-session IDs were:

Cycle	Researcher	Worker	Auditor
44	2c02d302-841f-41a7-b137-b0ebba5ba124	ca2ffce6-0b01-4b7b-bd7c-02e83530dada	6a8a5c72-28dd-4039-8746-332a852f907e
45	0fbe1028-73bd-4013-ba96-8e1a86b3d44f	983e9eb0-d654-44f0-92da-2f23e1c29079	065db097-5895-470e-a6da-71d16016ffbb
46	c01da8eb-3b85-4aa3-a71b-373f6f066ac2	14871cea-47d3-42b6-83dc-89039e3ca94c	dbf0dc18-fee6-4dd0-a1bf-60666ec5eb99

Full native transcripts for these sessions could not be fetched in this environment because session-search/session-fetch tools were not available. That is a record gap. The report therefore relies on the supplied audit and local project artifacts.

No report markdown was written to disk by the reporter. Per the orchestration protocol, the report content is supplied in this output block for the harness to write and render.

Findings

Validated No-Op Maintenance Decision

The central result for cycles 44-46 is a validated no-op. The audit states that no concrete maintenance defect, reproducibility failure, public-facing issue, or qualifying scientific evidence was supplied. Because the trigger predicate was absent, the worker correctly avoided build/run action, artifact regeneration, file edits, and ledger promotion.

This is a maintenance result, not a new scientific result. It does not add a prediction, validation, falsification, source, benchmark, figure, schema change, Atlas change, or track-local output.

Track Reopening Eligibility

No track reopening predicate was met. The six final track outcomes remain preserved:

Track	Preserved status	Meaning
Track 1 Reticulation Atlas	sidecar_readiness_uncontrolled	Reticulation-sidecar readiness remains insufficient for master promotion because accepted-key recovery and control requirements remain unresolved.
Track 2 Ghost Hyperedges	H2_remains_not_supported_or_data_limited	Held-out anachronism recovery remains unsupported or data-limited under the campaign validation rule.
Track 3 Convergence Pressure	confound_limited	Convergence-pressure evidence remains limited by family-size and sampling-density confounds.
Track 4 Domestication Hypergraph	still_data_limited	Crop-wild-relative and climate-envelope evidence remains insufficient for recommendation-like substitution claims.
Track 5 Chemodiversity Predictor	H5_remains_source_biased	Temporal phytochemistry validation remains source-biased and insufficient for promotion.
Track 6 Foundation Model Probe	environment_limited_untested	The static benchmark and deterministic harness remain deliverables, but no qualifying local/open model-response evaluation was promoted.

These statuses are reported as preserved from prior closure work. Cycles 44-46 did not produce new evidence that would change any track status.

Master Ledgers

The master ledgers remained non-promoted:

Ledger	Count	Status
prediction_ledger.tsv	1 line	Header-only
speculation_ledger.tsv	1 line	Header-only
promise_ledger.jsonl	237 lines	Append-only event ledger

The reporter also read the first rows of the prediction and speculation ledgers. Each file contains only its header. This supports the audit statement that no master prediction or speculation rows were promoted.

Validators And Warnings

The supplied audit reports these checks:

Check	Result
<code>wc -l prediction_ledger.tsv</code>	1, 1, and 237
<code>speculation_ledger.tsv promise_ledger.jsonl</code>	
<code>python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace</code>	exit 0; events: 237, plan milestones: 46
<code>python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace</code>	exit 0 with inherited root-layout warnings only

The inherited backlog includes the known immutable ledger-line-85 exception, historical non-canonical artifact paths, plan milestone warnings, orphan `reports/final/*` artifacts, and missing legacy fork/report/manager/cycle artifacts. The audit characterizes these as inherited, nonblocking warnings rather than new repair predicates.

The reporter did not rerun the validator suite. The validator results above are reported from the supplied audit.

Artifact Changes

The supplied audit states that no files were edited, no artifacts were rebuilt, and no ledger rows were promoted during the worker/auditor cycle work.

For this reporter pass, `MANIFEST.md` was updated as required for `report_cycles_44-46`. The update preserved the existing `## Key Files` section verbatim, changed the report window from cycles 41-43 to cycles 44-46, updated the promise-ledger count from 236 to 237, recorded the 46 plan milestones reported by `promise_check`, and restated the no-op terminal-maintenance posture.

No cycle-specific figures were found under `reports/cycles/` for this report window, and no figures are embedded in this report.

Discussion

Cycles 44-46 are a terminal maintenance interval. Their significance is that the campaign did not drift after closure. No track-local hypothesis was promoted, no master-ledger row was added, no validator failure appeared, and no public or handoff artifact was changed by the worker/auditor cycle work.

The audit guidance remains the controlling future-work rule: continue terminal maintenance unless future input supplies a concrete maintenance defect, reproducibility failure, public-facing clarity/accessibility or broken-link issue, or qualifying scientific evidence with auditable validation and ablation controls. Without that predicate, reopening tracks, changing schema or evidence tables, regenerating continuity-only reports, or promoting ledger rows would add process churn rather than scientific progress.

The original campaign success criterion remains unmet. PhytoGraph produced substantial infrastructure, validation, falsification, public communication, and handoff artifacts, but it did not produce at least one validated prediction per track. The conservative closure state is therefore preserved.

Open Questions

The remaining questions are maintenance gates rather than active research questions:

- Will a future cycle supply a concrete defect, reproducibility failure, public-facing issue, or qualifying scientific evidence?
- If new evidence is supplied, will it satisfy the established reopening predicates before any track status or master ledger changes?
- Can inherited validator warnings remain documented as historical backlog unless a new validator run reports a new error?
- Can public-facing maintenance remain limited to clarity, accessibility, link repair, provenance hygiene, and reproducibility?

References

No external bibliographic references are cited for cycles 44-46. REFERENCES.md was read for citation continuity, but this report cites only local project artifacts and the supplied audit record.

Appendix: Implementation Details

Source Inventory

Source	Date / cycle	Contents	Timeline role
Supplied audit report	Cycles 44-46	Validation summary, no-op decision, ledger counts, validator outcomes, sub-topic assessment, and future guidance	Primary validation source
Cycle 44 sessions	Cycle 44	Session IDs supplied; full transcripts unavailable locally	Source gap
Cycle 45 sessions	Cycle 45	Session IDs supplied; full transcripts unavailable locally	Source gap
Cycle 46 sessions	Cycle 46	Session IDs supplied; full transcripts unavailable locally	Source gap
reports/cycles/report_cycles_41-43.md	Prior reporting window	Previous terminal-maintenance report	Immediate continuity source
reports/final_campaign_handoff_manifest.md	Final handoff	Canonical closure state, track statuses, warning posture, and reopening conditions	Closure-boundary source
prediction_ledger.tsv	Current workspace	Header-only master prediction ledger	Non-promotion evidence
speculation_ledger.tsv	Current workspace	Header-only master speculation ledger	Non-promotion evidence
promise_ledger.jsonl	Current workspace	237 append-only events	Ledger-state source

Source	Date / cycle	Contents	Timeline role
MANIFEST.md	Current workspace	Updated snapshot for future researcher/worker turns	Reporter-maintained artifact

Code Organization

No code was added or modified as part of the worker/auditor cycle work. Existing referenced maintenance code remains:

File	Lines	Purpose
scripts/build_taxonomy_results_site_assets.py	428	Builds the public taxonomy-results review site data, evidence tables, and figure assets from the final six-track status table.
tests/test_taxonomy_results_site_public_text.py	130	Verifies required site files, figure references, route labels, public-language boundary, six status codes, and header-only master ledgers.
reports/reopen/scripts/build_final_free_tier_closure_synthesis.py	197	Builds the final six-track free-tier status table, closure report, and summary figure consumed by the handoff and site artifacts.
tests/test_final_free_tier_closure_synthesis.py	109	Verifies the final six-track status table, root/reopen propagation, figure presence, and header-only master ledgers.

File Counts

File	Count
prediction_ledger.tsv	1 line
speculation_ledger.tsv	1 line
promise_ledger.jsonl	237 lines
MANIFEST.md after reporter update	119 lines
REFERENCES.md	157 lines
reports/final_campaign_handoff_manifest.md	76 lines
reports/taxonomy_results_site_closure_note.md	15 lines
reports/taxonomy_results_site_qa.md	40 lines
reports/cycles/report_cycles_41-43.md	175 lines

Test Results

The reporter did not rerun the validator suite. The supplied audit reports:

- `wc -l prediction_ledger.tsv speculation_ledger.tsv promise_ledger.jsonl`: 1, 1, and 237.
- `python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace`: exit 0, events: 237, plan milestones: 46, inherited warnings only.
- `python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace`: exit 0 with inherited root-layout warnings only.

The reporter performed local line-count and header checks for the master ledgers and confirmed both master ledgers remain header-only.

Cross-Reference Map

Origin	Consumer	Flow
Supplied cycles 44-46 audit report	This periodic report	Establishes the validated no-op terminal-maintenance decision.
reports/cycles/report_cycles_41-43.md	This periodic report	Provides immediate prior terminal-maintenance context.
reports/final_campaign_handoff_manifest.md	Future maintainer turns	Preserves canonical closure artifacts, final track outcomes, warning posture, and reopening condition.
prediction_ledger.tsv and speculation_ledger.tsv	Closure records and future cycles	Header-only state confirms no master prediction or speculation promotion.
promise_check and org_check audit results	Audit decision	Confirm validator success with inherited nonblocking warnings only.
MANIFEST.md	Future researcher/worker turns	Updated to summarize cycles 44-46 while preserving ## Key Files verbatim.

Record Gaps

Full native transcripts for the supplied researcher, worker, and auditor session IDs were not available because session-search/session-fetch tools were not available in this environment.

No pre-existing local reports/cycles/report_cycles_44-46.* artifact was found before this final output. That is expected under the reporter protocol: the orchestrator writes the report markdown and renders the PDF from this output block.

Coherence Review

One coherence pass was completed. The report defines the terminal-maintenance posture, distinguishes no-op validation from scientific progress, states the master-ledger non-promotion rule before applying it, reports inherited warnings as inherited rather than new defects, and marks unavailable session transcripts as a record gap.